

duplicate elements, we specifically call them *k-combinations with repetition*.

Keep in mind that, both with and without repetition, you can think of permutation as a certain arrangement of all elements in a set, while a combination is an orderless selection of certain elements from a set. Permutations have an ordering and use all the elements from a set, while combinations don't have an ordering and use any number of elements from a set. To get a better idea of these terms, [Table 6-1](#) shows the difference between permutations and combinations, with and without repetition, of the set {A, B, C}.

Table 6-1: All Possible Permutations and Combinations, with and without Repetition, of the Set {A, B, C}

	Permutations	Combinations
Without repetition	ABC, ACB, BAC, BCA, CAB	(None), A, B, C, AB, AC, BC, ABC
With repetition	AAA, AAB, AAC, ABA, ABB, ABC, ACA, ACB, ACC, BAA, BAB, BAC, BBA, BBB, BBC, BCA, BCB, BCC, CAA, CAB, CAC, CBA, CBB, CBC, CCA, CCB, CCC	(None), A, B, C, AA, AB, AC, BB, BC, CC, AAA, AAB, AAC, ABB, ABC, ACC, BBB, BBC, BCC, CCC

It's surprising how quickly the number of permutations and combinations grows as we add elements to a set. This *combinatorial explosion* is captured by the formulas in [Table 6-2](#). For example, a set of 10 elements has 10!, or 3,628,800, possible permutations, but a set of twice as many elements has 20!, or 2,432,902,008,176,640,000, permutations.

Table 6-2: Calculating the Number of Possible Permutations and Combinations, with and without Repetition, of a Set of n Elements

	Permutations	Combinations
Without repetition	$n!$	2^n
With repetition	n^n	2^n choose n , or $(2n)! / (n!)^2$

Note that permutations without repetition are always the same size as the set. For example, the permutations of {A, B, C} are always three letters long: ABC, ACB, BAC, and so forth. However, permutations with repetition can be of any length. [Table 6-1](#) shows the three-letter